

Getting Started with Mininet-WiFi

■ Outcomes

- 2.1 Monitor and analyze WiFi traffic
- 2.2 Create a managed WiFi network

■ Background

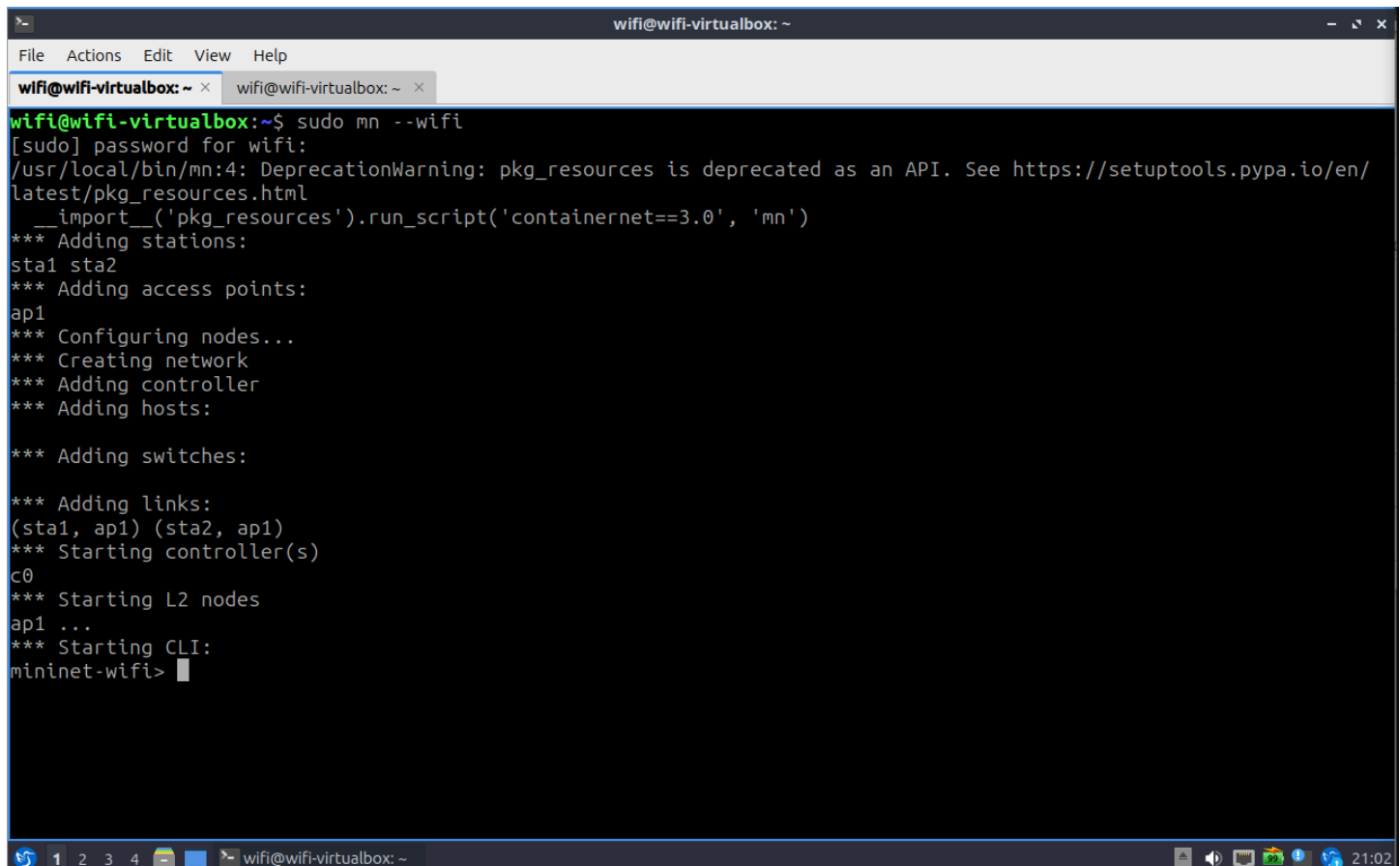
In this lab you will use a Mininet-WiFi VM image to run a wireless SDN network emulator and discover the properties of a basic WiFi network.

■ Running the VM

If you don't already have it installed, download [VirtualBox](#) for your platform. You will also need to download the [OVA file for the Mininet-WiFi VM](#) (I know it's huge, but we'll reuse it). *Be sure to use your university Google Workspace account and not your personal account.* Once downloaded, import the file into VirtualBox (File->Import Appliance). Start the machine and after it boots you will see the Lubuntu Desktop.

■ Running Mininet-WiFi

Click the menu in the bottom left and open a terminal (System Tools->QTerminal). In the terminal run `sudo mn --wifi` (the wifi user password is wifi).



```
wifi@wifi-virtualbox: ~$ sudo mn --wifi
[sudo] password for wifi:
/usr/local/bin/mn:4: DeprecationWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html
__import__('pkg_resources').run_script('containernet==3.0', 'mn')
*** Adding stations:
sta1 sta2
*** Adding access points:
ap1
*** Configuring nodes...
*** Creating network
*** Adding controller
*** Adding hosts:

*** Adding switches:

*** Adding links:
(sta1, ap1) (sta2, ap1)
*** Starting controller(s)
c0
*** Starting L2 nodes
ap1 ...
*** Starting CLI:
mininet-wifi>
```

By default Mininet-WiFi creates three nodes: Two stations and one access point. This is a very basic WiFi network utilizing managed mode. From the Mininet-WiFi command line you can run Linux commands as any station or access point.

For example, try running `sta1 ping -c sta2`. You should see three ping responses from sta2.

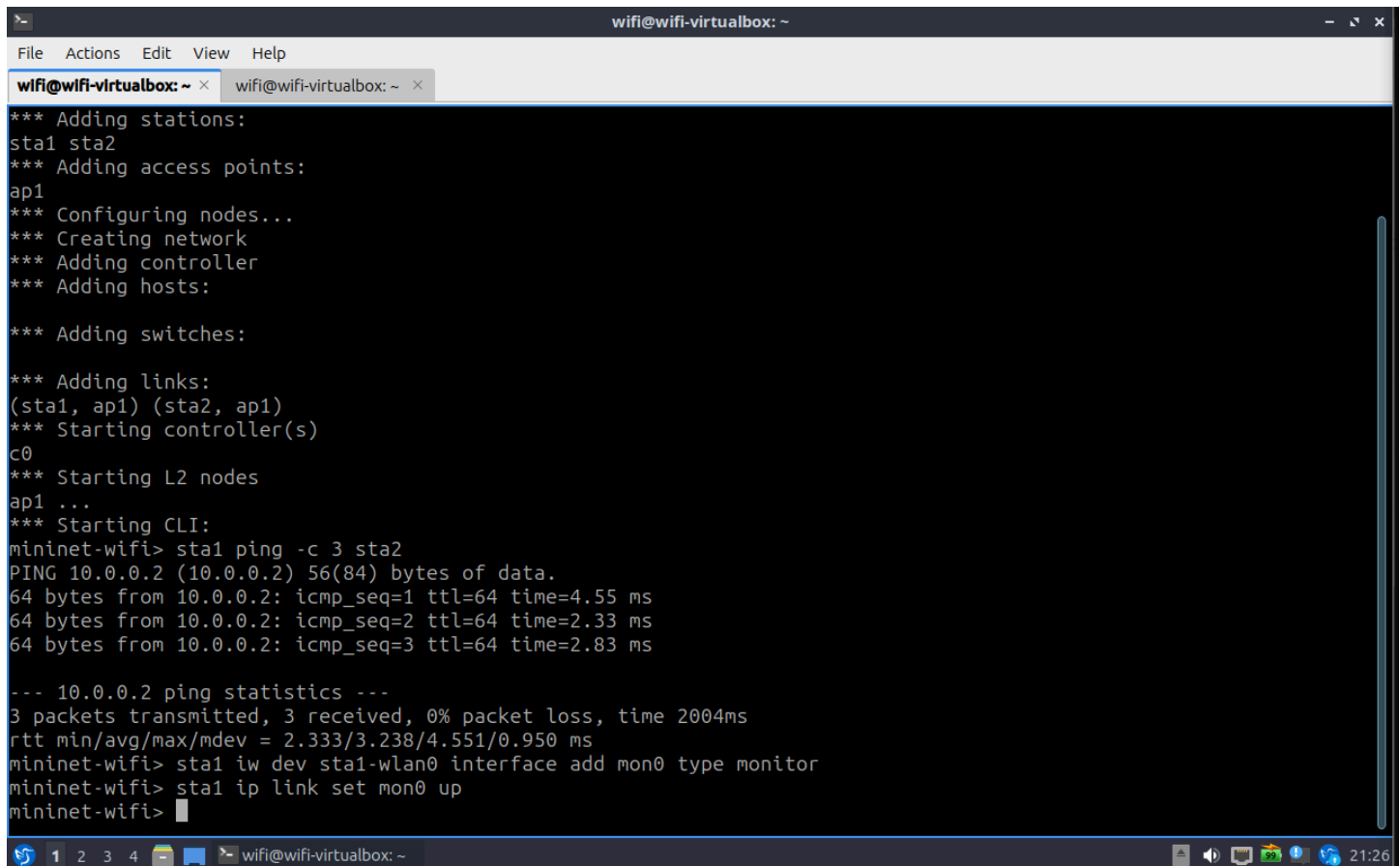
■ Adding a Monitor Interface to sta1

If we want to see all of the traffic that a node in our network sees, we'll need to add a monitor interface. We can do this from the command line with the following commands:

```
sta1 iw dev sta1-wlan0 interface add mon0 type monitor
```

This basically says, "Run the iw command on sta1 and tell it that I want to add a monitor device to the sta1-wlan0 interface."

Next we run `sta1 ip link set mon0 up` which uses the ip command on sta1 to bring the monitor interface up.



```
wifi@wifi-virtualbox: ~
File Actions Edit View Help
wifi@wifi-virtualbox: ~ x wifi@wifi-virtualbox: ~ x
*** Adding stations:
sta1 sta2
*** Adding access points:
ap1
*** Configuring nodes...
*** Creating network
*** Adding controller
*** Adding hosts:

*** Adding switches:

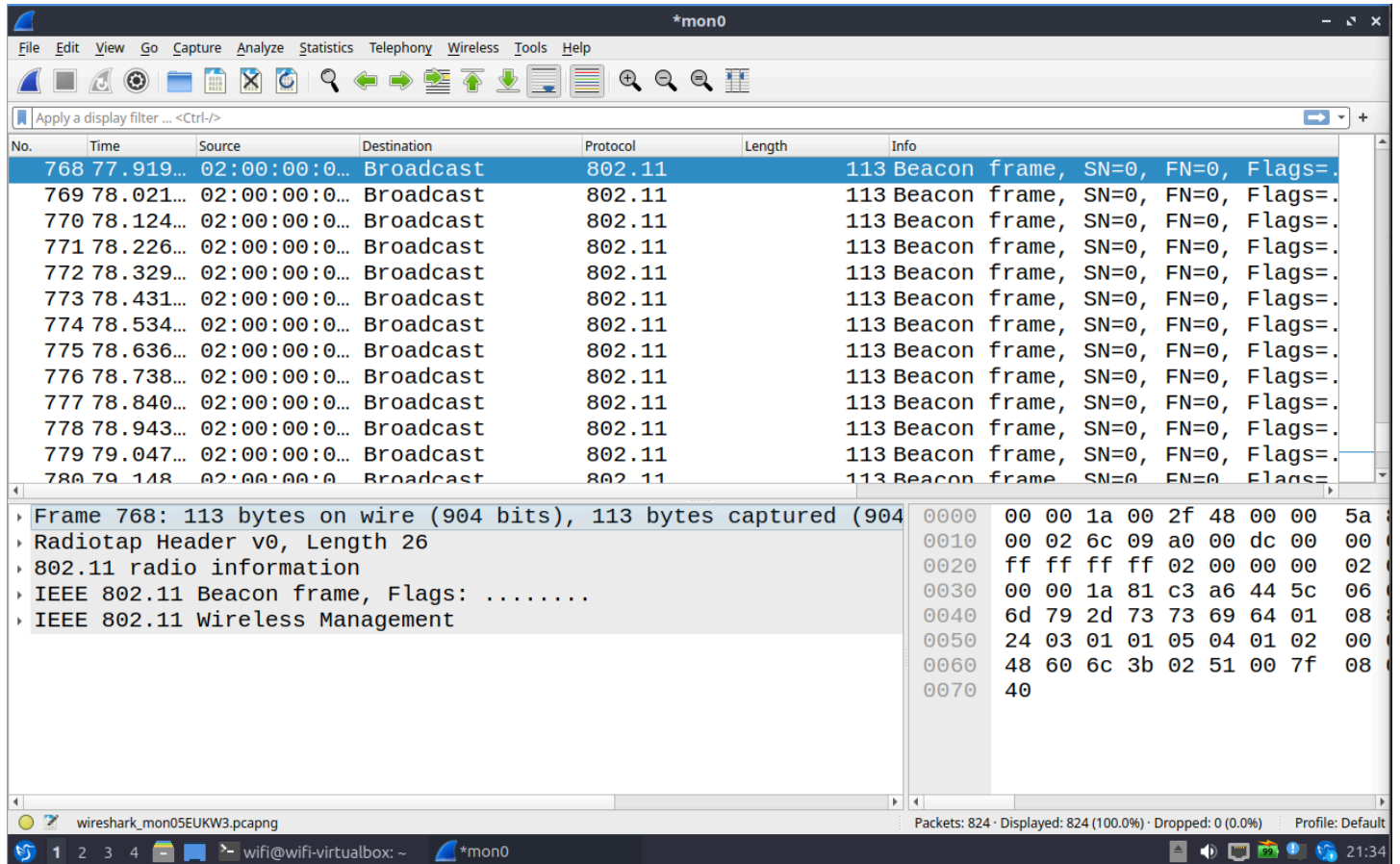
*** Adding links:
(sta1, ap1) (sta2, ap1)
*** Starting controller(s)
c0
*** Starting L2 nodes
ap1 ...
*** Starting CLI:
mininet-wifi> sta1 ping -c 3 sta2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=4.55 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=2.33 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=2.83 ms

--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 2.333/3.238/4.551/0.950 ms
mininet-wifi> sta1 iw dev sta1-wlan0 interface add mon0 type monitor
mininet-wifi> sta1 ip link set mon0 up
mininet-wifi> 
```

■ Running Wireshark on sta1

Now try running `sta1 wireshark &` This should start the Wireshark Network Analyzer as if it is running on sta1.

Double-click on mon0 to start capturing on that interface. You can hit the red stop square in the upper-left corner after you grab a few frames (you probably have more than a few at this point).



Questions

Use the information in the Wireshark packets to answer the following questions for this lab:

1. What kind of frames are you seeing?
2. What is the MAC address of the AP (BSS Id)?
3. What is the SSID of the WiFi network?

When you are done you can close Wireshark, type exit in the Mininet-WiFi command line, and Shutdown the machine from the main menu.